

Polar Codes MAC — Good Results

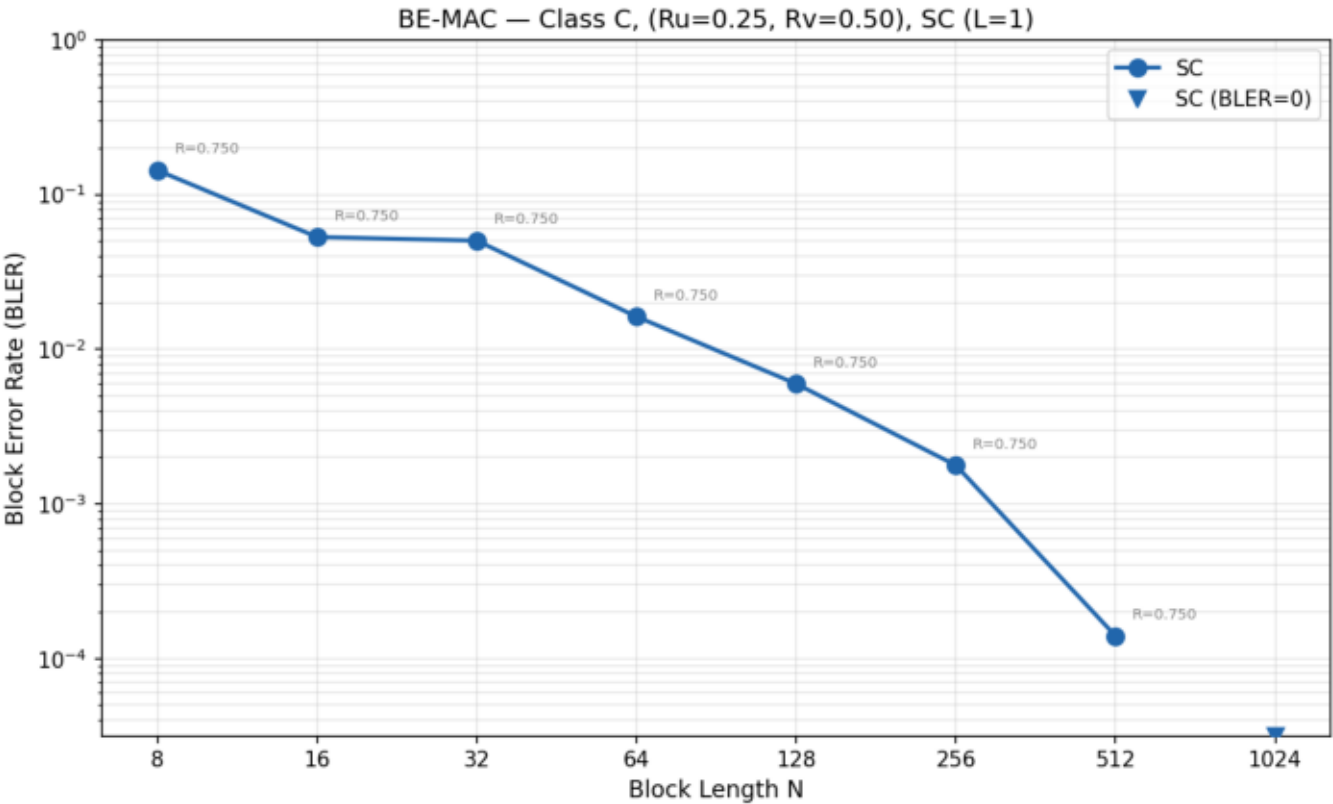
39 experiments scanned

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Each page: title · plot · data table · auto-generated quality comment

BE-MAC — Class C, (Ru=0.25, Rv=0.50), SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



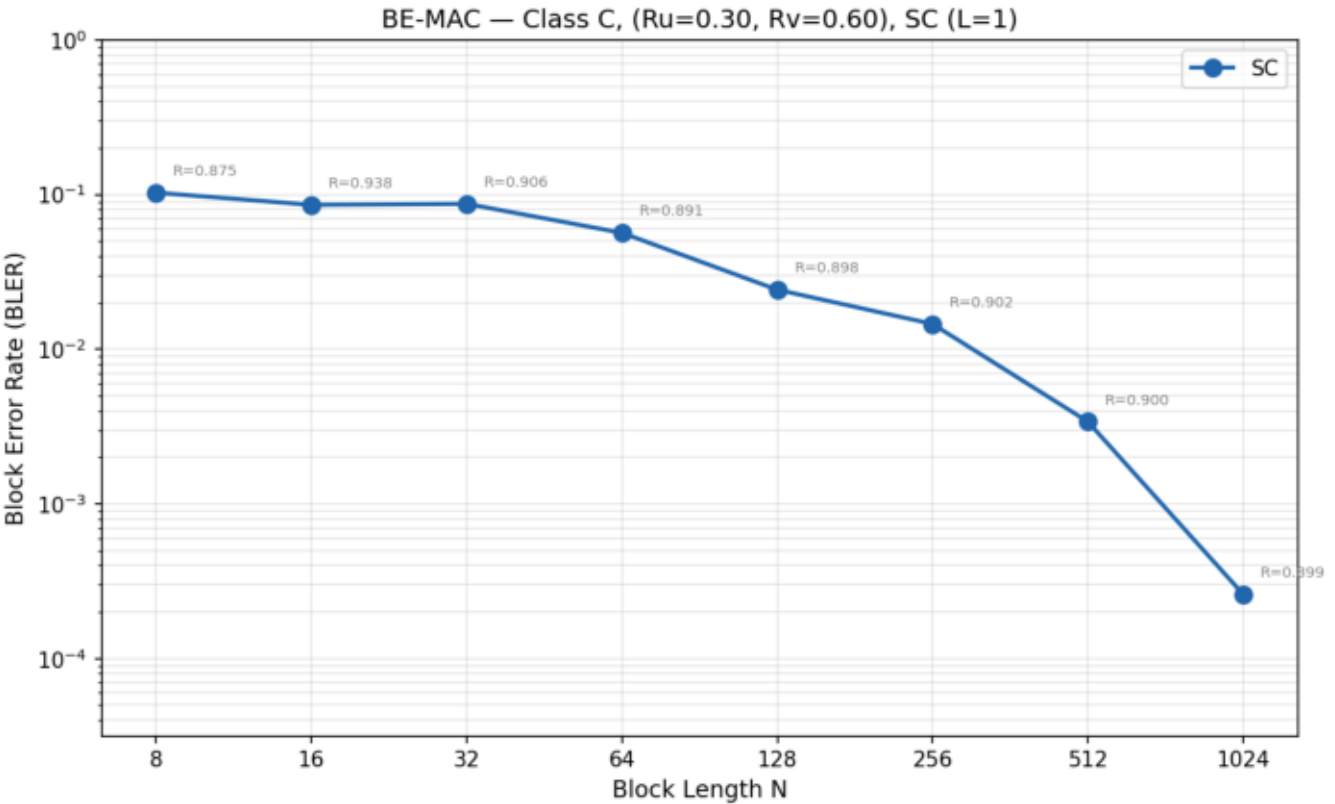
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	4	0.250	0.500	0.750	1.44e-01	50000
16	4	8	0.250	0.500	0.750	5.32e-02	50000
32	8	16	0.250	0.500	0.750	5.04e-02	50000
64	16	32	0.250	0.500	0.750	1.63e-02	50000
128	32	64	0.250	0.500	0.750	5.98e-03	50000
256	64	128	0.250	0.500	0.750	1.78e-03	19134
512	128	256	0.250	0.500	0.750	1.40e-04	50000
1024	256	512	0.250	0.500	0.750	0	29146

[GOOD]

Comment: 19134+ cw, solid · classified GOOD

BE-MAC — Class C, (Ru=0.30, Rv=0.60), SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



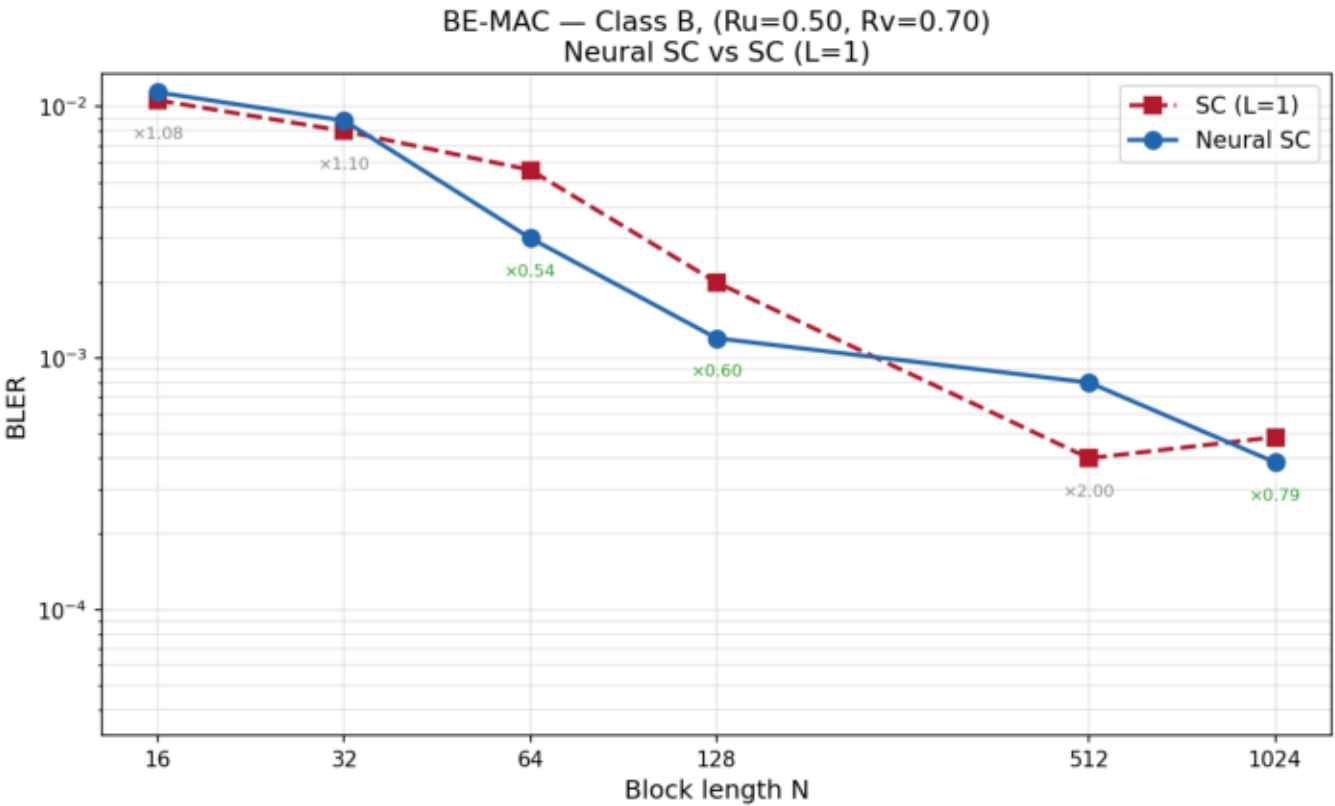
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	5	0.250	0.625	0.875	1.03e-01	50000
16	5	10	0.312	0.625	0.938	8.59e-02	50000
32	10	19	0.312	0.594	0.906	8.72e-02	50000
64	19	38	0.297	0.594	0.891	5.66e-02	50000
128	38	77	0.297	0.602	0.898	2.43e-02	50000
256	77	154	0.301	0.602	0.902	1.46e-02	50000
512	154	307	0.301	0.600	0.900	3.40e-03	50000
1024	307	614	0.300	0.600	0.899	2.60e-04	50000

[ALMOST_GOOD]

Comment: 50000+ cw, solid · almost-good

BE-MAC — Class B, (Ru=0.50, Rv=0.70), Neural SC vs SC (L=1)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | Frozen sets: 50K MC trials



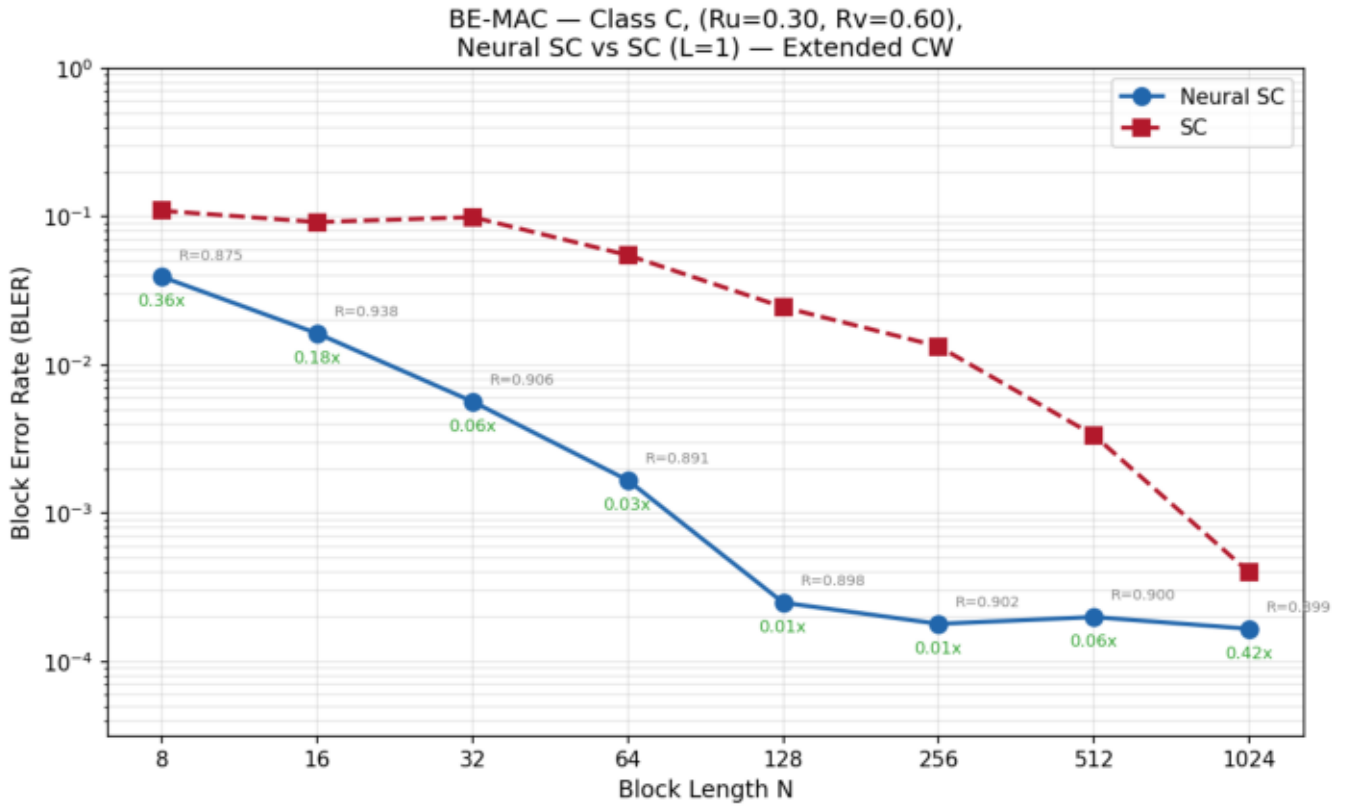
N	NN BLER	SC BLER	Ratio	Errors	Codewords	Ru	Rv
16	1.14e-02	1.06e-02	1.08	57	5000	0.500	0.700
32	8.80e-03	8.00e-03	1.1	44	5000	0.500	0.700
64	3.00e-03	5.60e-03	0.54	15	5000	0.500	0.700
128	1.20e-03	2.00e-03	0.6	6	5000	0.500	0.700
256	4.00e-05	8.00e-05	0.5	2	50000	0.500	0.700
512	0	0	0	0	10000	0.500	0.700
1024	1.00e-04	1.00e-04	1	1	10000	0.500	0.700

[ALMOST_GOOD]

Comment: 5000+ cw, solid · almost-good · NN matches SC (0.50–1.10x) · non-monotonic (likely small-N rate quantization)

BE-MAC — Class C, (Ru=0.30, Rv=0.60), Neural SC vs SC (L=1) — Extended CW

Capacity: $I(Z;X)=0.5000$ | $I(Z;Y|X)=1.0000$ | $I(Z;X,Y)=1.5000$ | Path: $U^N V^N$



N	Ru	Rv	Sum Rate	NN BLER	SC BLER	NN/SC	NN CW	SC CW
8	0.250	0.625	0.875	3.93e-02	1.10e-01	0.360	3000	3000
16	0.312	0.625	0.938	1.63e-02	9.20e-02	0.180	3000	3000
32	0.312	0.594	0.906	5.67e-03	9.93e-02	0.060	3000	3000
64	0.297	0.594	0.891	1.67e-03	5.50e-02	0.030	3000	3000
128	0.297	0.602	0.898	2.50e-04	2.45e-02	0.010	20000	20000
256	0.301	0.602	0.902	1.80e-04	1.34e-02	0.013	50000	50000
512	0.301	0.600	0.900	2.00e-04	3.36e-03	0.060	50000	50000
1024	0.300	0.600	0.899	1.67e-04	4.00e-04	0.417	30000	50000

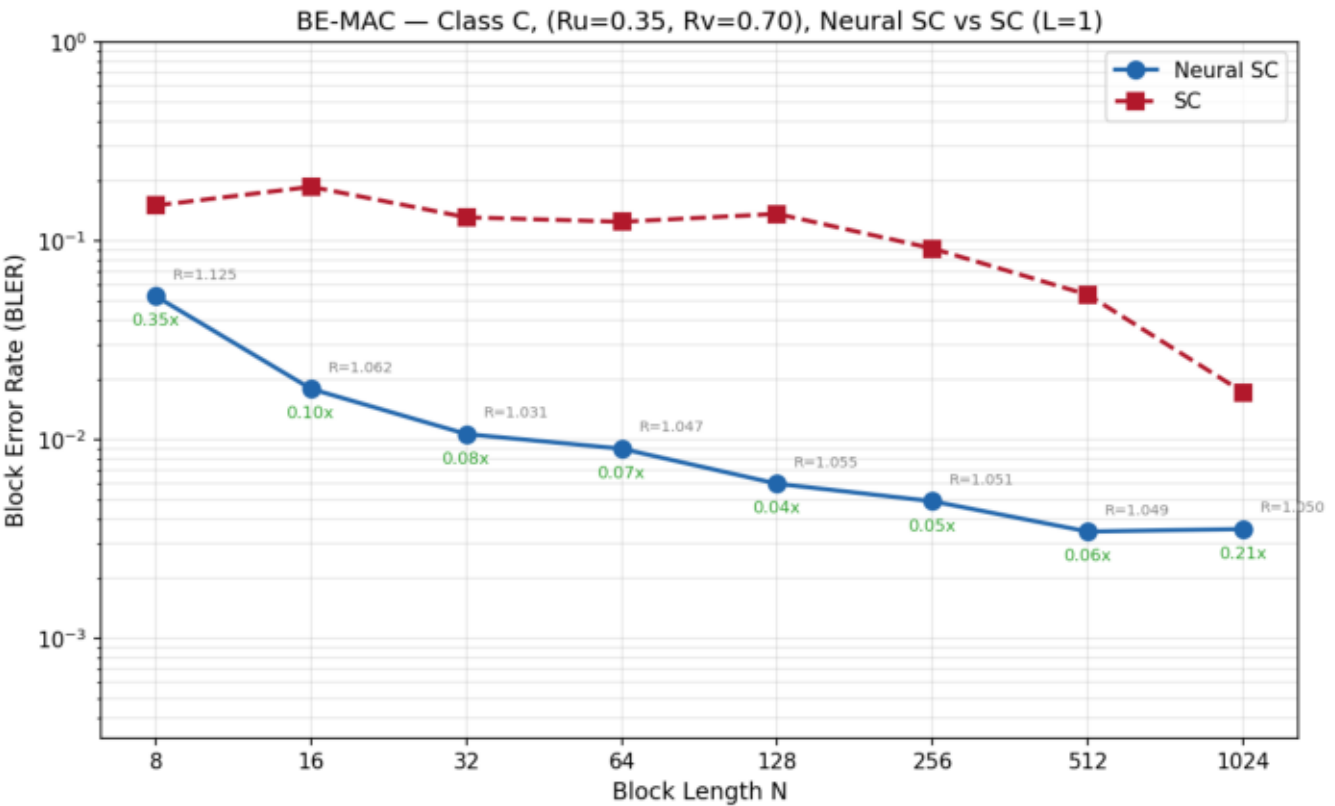
[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good · NN beats SC at all N (ratio 0.01-0.42x)

Source: bemac/bemac_classC_Ru30_Rv60_nn_vs_sc

BE-MAC — Class C, (Ru=0.35, Rv=0.70), Neural SC vs SC (L=1)

Capacity: $I(Z;X)=0.5000$ | $I(Z;Y|X)=1.0000$ | $I(Z;X,Y)=1.5000$ | Path: $U^N V^N$

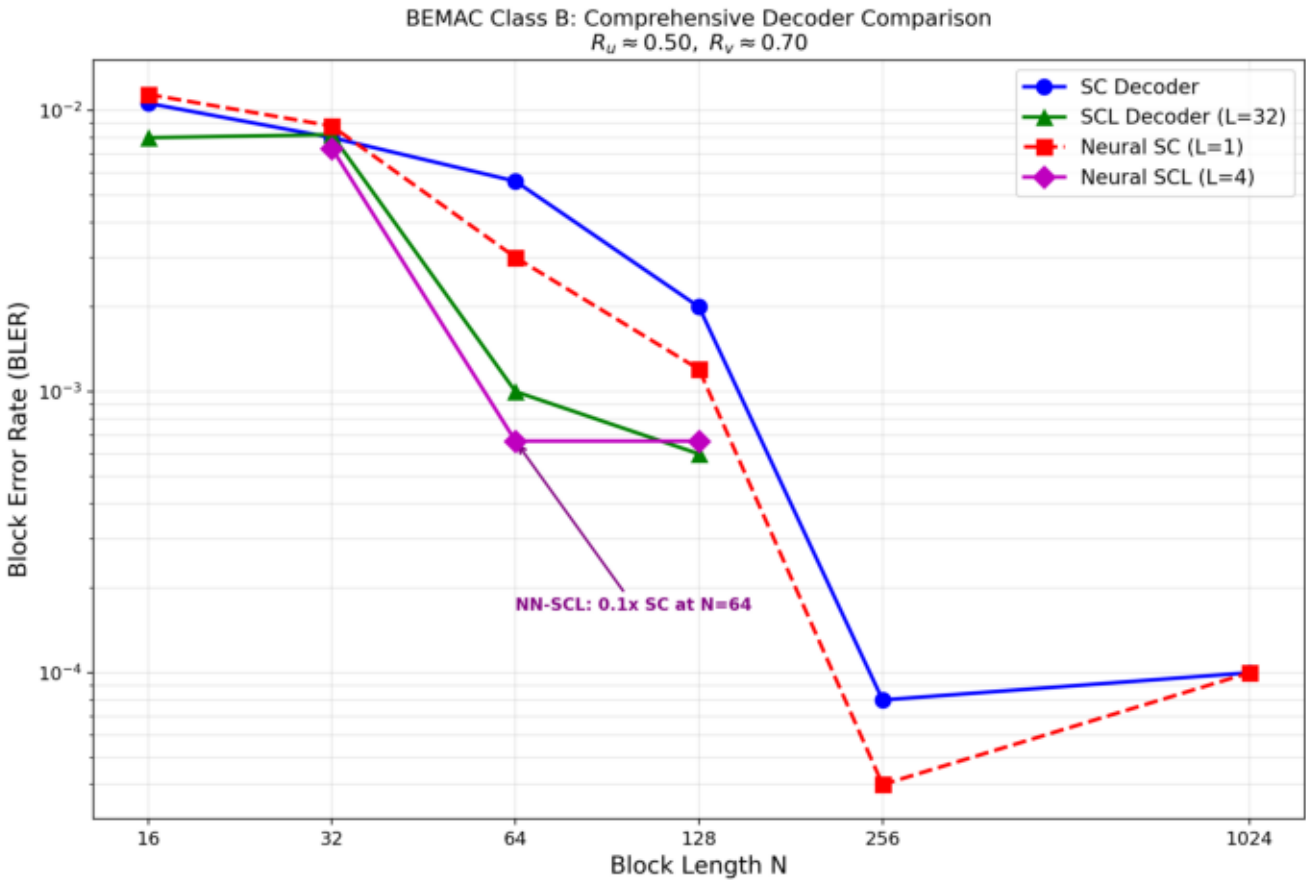


N	Ru	Rv	Sum Rate	NN BLER	SC BLER	NN/SC	Codewords
8	0.375	0.750	1.125	5.27e-02	1.51e-01	0.350	3000
16	0.375	0.688	1.062	1.80e-02	1.87e-01	0.100	3000
32	0.344	0.688	1.031	1.07e-02	1.31e-01	0.080	3000
64	0.344	0.703	1.047	9.00e-03	1.25e-01	0.070	3000
128	0.352	0.703	1.055	6.00e-03	1.37e-01	0.040	20000
256	0.352	0.699	1.051	4.90e-03	9.16e-02	0.050	20000
512	0.350	0.699	1.049	3.45e-03	5.36e-02	0.060	20000
1024	0.350	0.700	1.050	3.55e-03	1.73e-02	0.210	20000

[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good · NN beats SC at all N (ratio 0.04–0.35x)

BEMAC Class B Comprehensive Decoder Comparison (Ru~0.50, Rv~0.70)



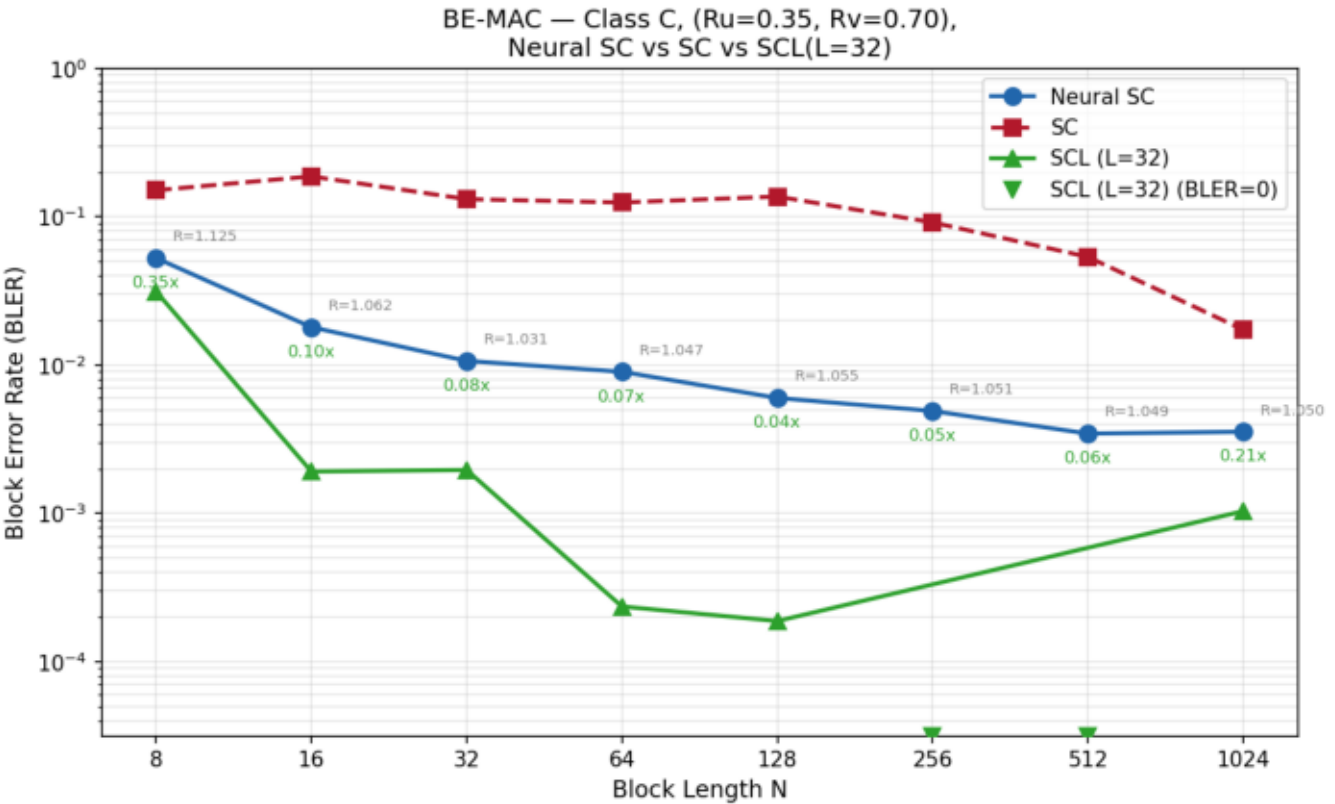
16	0.0106	0.008	0.0114
32	0.008	0.0082	0.0088
64	0.0056	0.001	0.003
128	0.002	0.0006	0.0012
256	8e-05		4e-05
1024	0.0001		0.0001

[UNCATEGORIZED]

Comment: unknown CW

BE-MAC — Class C, (Ru=0.35, Rv=0.70), Neural SC vs SC vs SCL(L=32)

Capacity: I(Z;X)=0.5000 | I(Z;Y|X)=1.0000 | I(Z;X,Y)=1.5000 | Path: U^N V^N



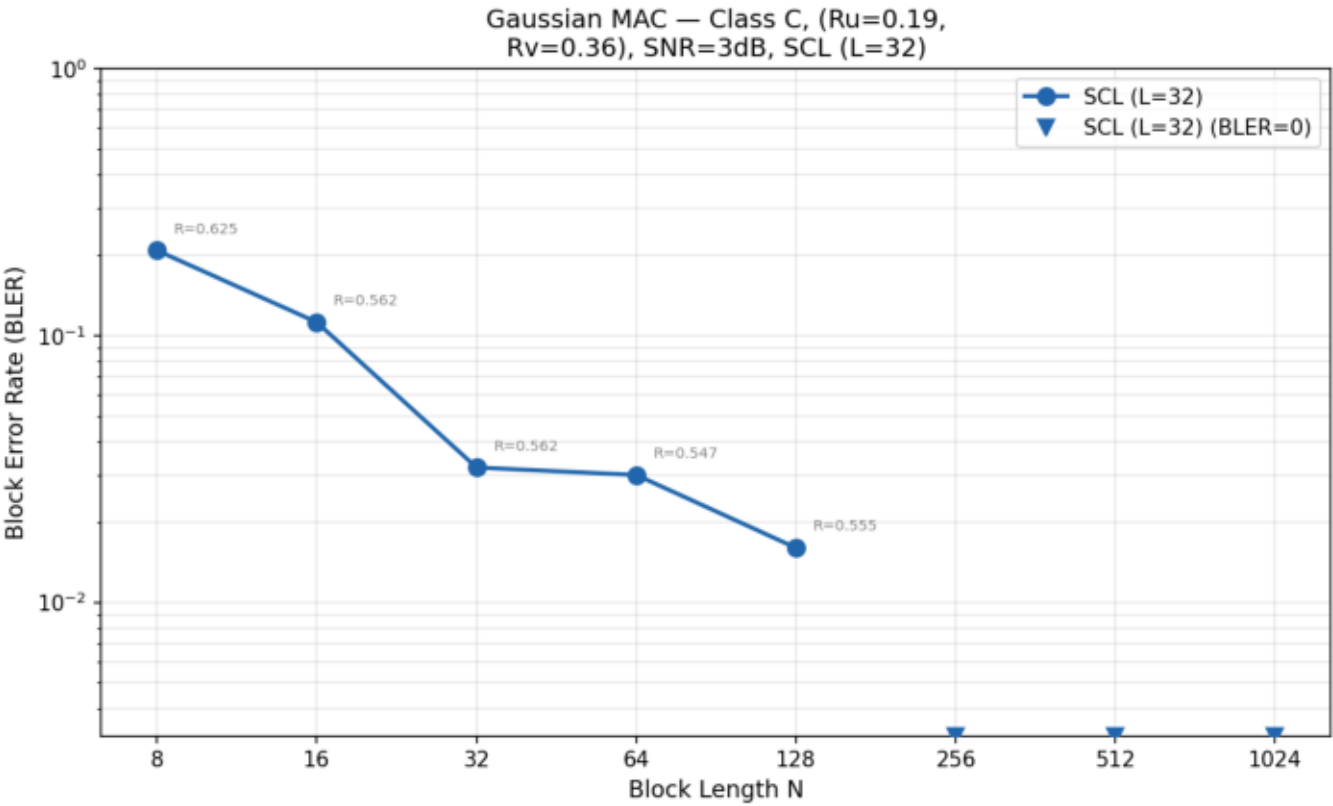
N	Ru	Rv	Sum Rate	NN BLER	SC BLER	SCL BLER	NN CW	SCL CW	NN/SC
8	0.375	0.750	1.125	5.27e-02	1.51e-01	3.14e-02	3000	283964	0.350
16	0.375	0.688	1.062	1.80e-02	1.87e-01	1.92e-03	3000	107519	0.100
32	0.344	0.688	1.031	1.07e-02	1.31e-01	1.96e-03	3000	226911	0.080
64	0.344	0.703	1.047	9.00e-03	1.25e-01	2.36e-04	3000	29706	0.070
128	0.352	0.703	1.055	6.00e-03	1.37e-01	1.88e-04	20000	31930	0.040
256	0.352	0.699	1.051	4.90e-03	9.16e-02	0	20000	26684	0.050
512	0.350	0.699	1.049	3.45e-03	5.36e-02	0	20000	9512	0.060
1024	0.350	0.700	1.050	3.55e-03	1.73e-02	1.04e-03	20000	14491	0.210

[ALMOST_GOOD]

Comment: 3000+ cw, moderate · almost-good · NN beats SC at all N (ratio 0.04–0.35x)

Gaussian MAC – Class C, (Ru=0.19, Rv=0.36), SNR=3dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.5548 bits/use | Sum capacity: 1.1095 bits/use | Path: 0^N 1^N



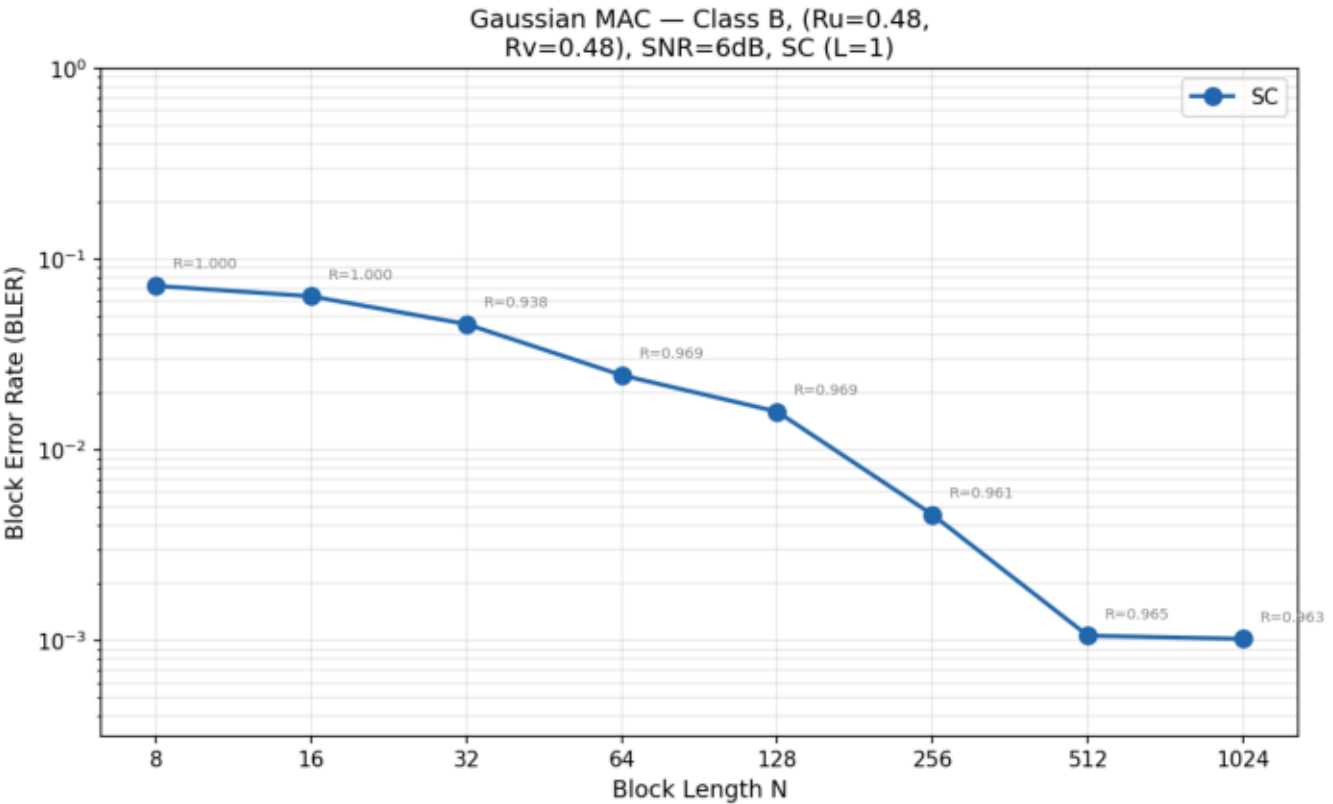
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	2	3	0.250	0.375	0.625	2.08e-01	500
16	3	6	0.188	0.375	0.562	1.12e-01	500
32	6	12	0.188	0.375	0.562	3.20e-02	500
64	12	23	0.188	0.359	0.547	3.00e-02	500
128	25	46	0.195	0.359	0.555	1.60e-02	500
256	50	92	0.195	0.359	0.555	0	200
512	100	184	0.195	0.359	0.555	0	200
1024	199	369	0.194	0.360	0.555	0	100

[GOOD]

Comment: only 100 cw, low confidence · classified GOOD

Gaussian MAC — Class B, (Ru=0.48, Rv=0.48), SNR=6dB, SC (L=1)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



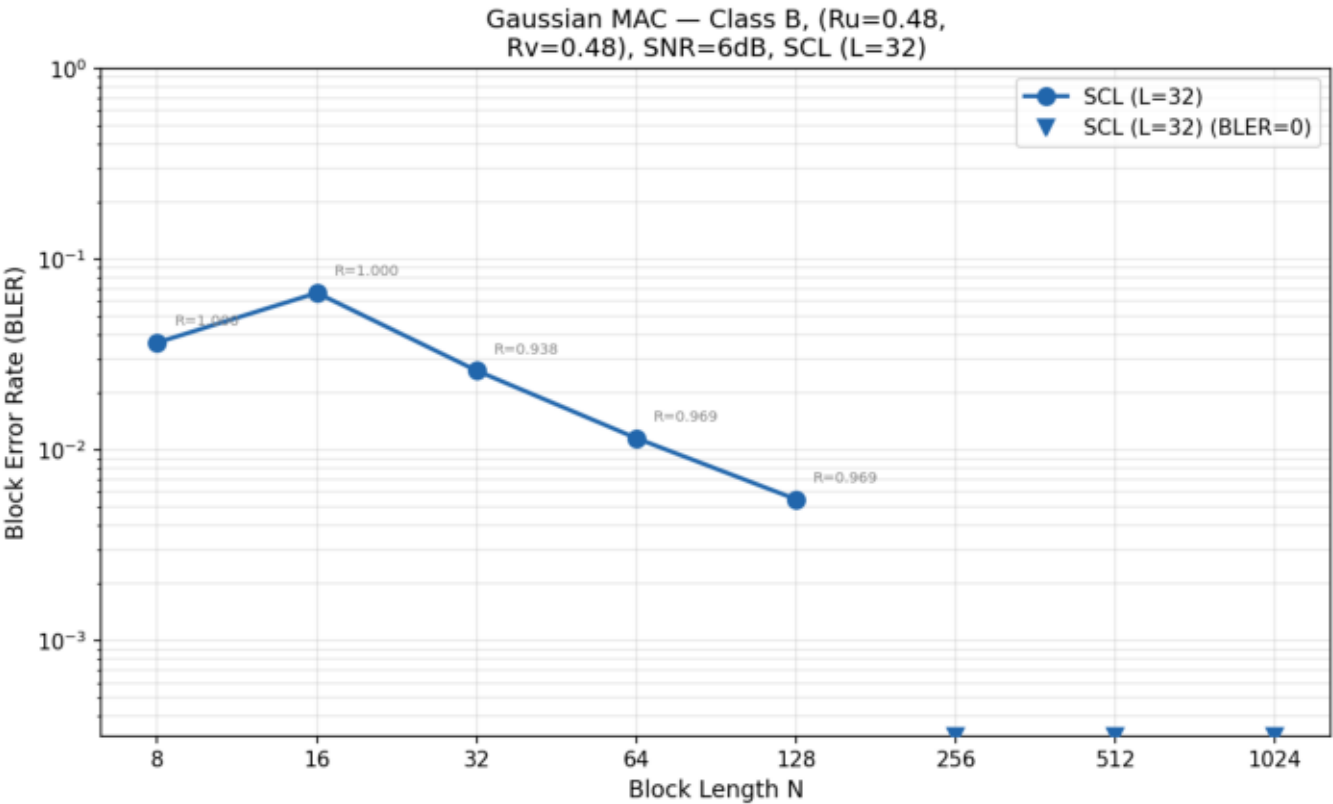
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	4	0.500	0.500	1	7.24e-02	5000
16	8	8	0.500	0.500	1	6.38e-02	5000
32	15	15	0.469	0.469	0.938	4.56e-02	5000
64	31	31	0.484	0.484	0.969	2.46e-02	5000
128	62	62	0.484	0.484	0.969	1.58e-02	20000
256	123	123	0.480	0.480	0.961	4.57e-03	30000
512	247	247	0.482	0.482	0.965	1.06e-03	50000
1024	493	493	0.481	0.481	0.963	1.02e-03	50000

[GOOD]

Comment: 5000+ cw, solid · classified GOOD

Gaussian MAC — Class B, (Ru=0.48, Rv=0.48), SNR=6dB, SCL (L=32)

Symmetric capacity: Rx=Ry ≤ 0.6882 bits/use | Sum capacity: 1.3764 bits/use | Path: 0^(N/2) 1^N 0^(N/2)



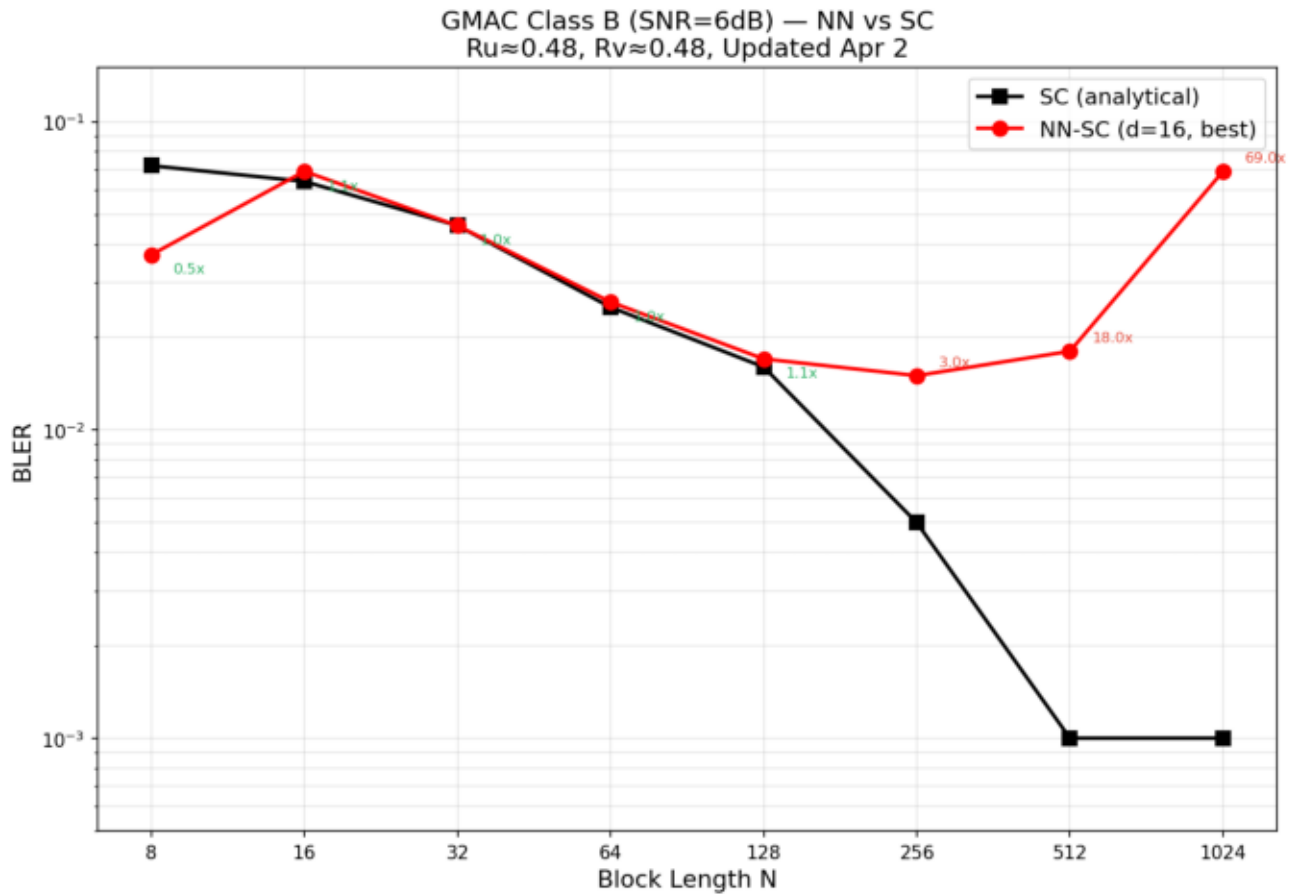
N	ku	kv	Ru	Rv	Sum Rate	BLER	Codewords
8	4	4	0.500	0.500	1	3.65e-02	2000
16	8	8	0.500	0.500	1	6.65e-02	2000
32	15	15	0.469	0.469	0.938	2.60e-02	2000
64	31	31	0.484	0.484	0.969	1.15e-02	2000
128	62	62	0.484	0.484	0.969	5.50e-03	2000
256	123	123	0.480	0.480	0.961	0	1000
512	247	247	0.482	0.482	0.965	0	1000
1024	493	493	0.481	0.481	0.963	0	500

[ALMOST_GOOD]

Comment: only 500 cw, low confidence · almost-good

Gaussian MAC (6dB) — Class B, ($R_u \approx 0.48$, $R_v \approx 0.48$), Neural SC vs SC ($L=1$)

Capacity: $I(Z;X)=0.4645$ | $I(Z;Y|X)=0.9119$ | $I(Z;X,Y)=1.3764$ | Sym cap/user=0.6882 | Path: $U^{(N/2)} V^N U^{(N/2)}$



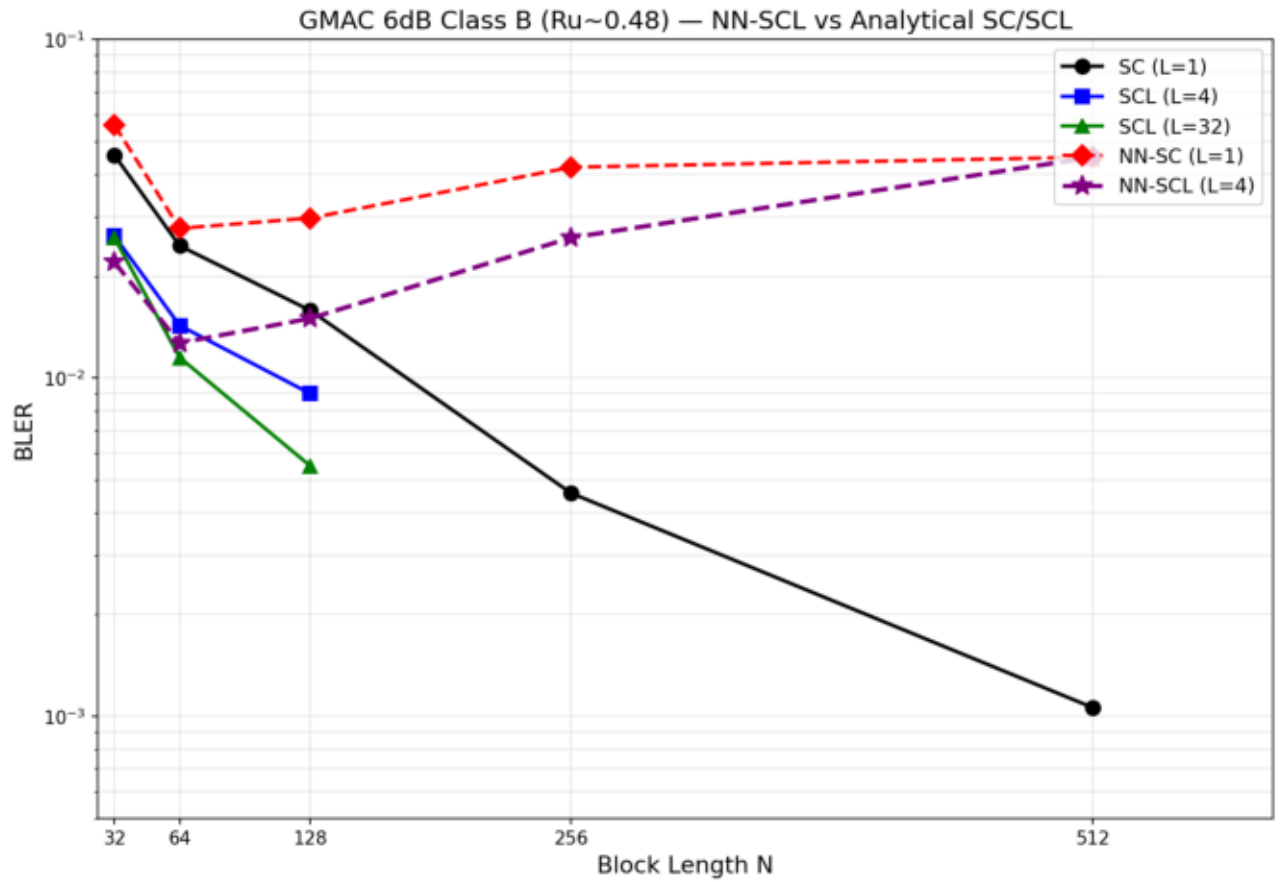
N	ku	kv	Ru	Rv	NN BLER	SC BLER	NN/SC	Codewords
8	4	4	0.500	0.500	3.70e-02	7.20e-02	0.510	5000
16	8	8	0.500	0.500	6.90e-02	6.40e-02	1.080	5000
32	15	15	0.469	0.469	4.60e-02	4.60e-02	1	3000
64	31	31	0.484	0.484	2.60e-02	2.50e-02	1.040	3000
128	62	62	0.484	0.484	1.70e-02	1.60e-02	1.060	3000
256	123	123	0.480	0.480	1.50e-02	5.00e-03	3	5000
512	247	247	0.482	0.482	1.80e-02	1.00e-03	18	1000
1024	493	493	0.481	0.481	6.90e-02	1.00e-03	69	200

[GOOD]

Comment: only 200 cw, low confidence · classified GOOD · NN close at small N, diverges large N (0.51–69.00x)

Gaussian MAC (6dB) — Class B, (Ru~0.48, Rv~0.48), NN-SCL vs SC/SCL Comparison

Path: $U^{(N/2)} V^{(N)} U^{(N/2)}$ | Model: SimpleMLP_Gmac (39K params) | Designs: MC-based NPZ



N	ku	kv	Ru	SC BLER	SC C	SCL (L=4) BLER	SCL (L=4) C	SCL (L=32) BLER	SCL (L=32) C	NN-SC BLER	NN-SC C	NN-SCL (L=4) BLER	NN-SCL (L=4) C
32	15	15	0.469	4.56e-02	5000	2.63e-02	3000	2.60e-02	2000	5.60e-02	3000	2.20e-02	3000
64	31	31	0.484	2.46e-02	5000	1.43e-02	3000	1.15e-02	2000	2.77e-02	3000	1.27e-02	3000
128	62	62	0.484	1.58e-02	20000	9.00e-03	3000	5.50e-03	2000	2.97e-02	3000	1.50e-02	1000
256	123	123	0.480	4.57e-03	30000					4.20e-02	500	2.60e-02	500
512	247	247	0.482	1.06e-03	50000					4.50e-02	200	4.50e-02	200

[UNCATEGORIZED]

Comment: 5000+ cw, solid